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ValSKA: Towards a UKSRC Validation Framework for SKA and Precursor Science

UKSRC MEMO

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Executive summary

The Square Kilometre Array (SKA) is being built to address frontier problems in astrophysics and cosmology, including observing Cosmic Dawn and the Epoch of Reionization and placing constraints on the Dark Energy equation of state with line intensity mapping. Achieving credible scientific inference in these regimes requires validation methods that can establish when complex interferometric workflows are reliable and where their limits lie. This memo sets out the technical rationale for ValSKA, an emerging UKSRC validation framework for SKA and precursor science workflows.

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List of abbreviations

Abbreviation	Description
SKA	Square Kilometre Array
UKSRC	United Kingdom SKA Regional Centre
CD	Cosmic Dawn
EoR	Epoch of Reionization
HERA	Hydrogen Epoch of Reionization Array

1 Introduction

The Square Kilometre Array (SKA[1]) is an international radio astronomy project constructing two major telescope arrays: SKA-Low in Australia, operating from 50 to 350 MHz, and SKA-Mid in South Africa, operating from 0.35 to 15.4 GHz. Together, these instruments are intended to address a broad range of questions in astrophysics, fundamental physics, cosmology, and particle astrophysics[2]. Among the most ambitious of these are science cases that seek to open new discovery space, including the detection and detailed characterisation of cosmological phenomena that have thus far eluded detection with current instruments. Examples in radio cosmology include line-intensity mapping observations of the 21-cm hyperfine line of neutral hydrogen. Measurements of this signal emitted in the Universe's first billion years after recombination have the potential to provide precision constraints on cosmological parameters during the Cosmic Dark Ages (e.g. Loeb & Zaldarriaga 2004; Mao et al. 2008) and to probe directly the initial stages of structure formation and characterise the properties of the first stars, proto-galaxies and accreting black holes during Cosmic Dawn (CD) and the Epoch of Reionization (EoR) (e.g. Furlanetto et al. 2006; Mesinger et al. 2013, 2014; Fialkov et al. 2014; Koopmans et al. 2015; Pritchard et al. 2015). At lower redshifts, line-intensity mapping observations will constrain the dark-energy equation of state (e.g. Square Kilometre Array Cosmology Science Working Group et al. 2020).

These regimes are scientifically rich but also methodologically demanding: the target signals are faint, obscured by bright foreground emission, and must be disentangled from complex instrumental systematics (e.g. Barry et al. 2016; Liu & Shaw 2020). Achieving credible inference in this setting requires structured validation: controlled tests, diagnostic analyses, and end-to-end studies that determine when scientific workflows are reliable and what level of instrument and sky characterisation they require.

For the purposes of this memo, we define validation as a multi-layered, explicitly scoped, simulation-driven process for verifying the robustness of scientific inference under controlled perturbations and independent implementations. Its goal is to establish justified confidence in a method by evaluating the problem space in which it works as intended, its modes of failure, and which uncertainties most strongly limit the scientific interpretation of the data.

The newly-established UKSRC science tooling programme is developing a framework to make validation tooling reusable across SKA and precursor instruments. Initial work has focussed on establishing a workflow for such studies. A validation campaign should define a concrete science question, construct realistic and reproducible simulations, perturb assumptions or systematics in a controlled fashion, and inspect how those changes propagate into scientific inference, by which we mean the use of data, models, and assumptions to draw scientific conclusions. The current demonstrator repository applies this pattern to evaluating the level of sensitivity of forward-modelling-based Bayesian 21-cm power-spectrum inference (e.g. Sims et al. 2016, 2019; Sims & Pober 2019; Burba et al. 2023; Sims et al. 2024) to uncertainty

[1] https://www.skao.int/en
[2] https://www.skao.int/en/science-users/science-working-groups

in the HERA primary-beam full width at half maximum (FWHM) associated with the antenna design and manufacturing process and electromagnetic environment of the array including mutual coupling (e.g. Fagnoni et al. 2021; Josaitis et al. 2022; Rath et al. 2025; O'Hara et al. 2025). Astrophysical and instrumental inputs to the demonstrator, the Bayesian null-test-based validation methodology used to evaluate the tolerance to beam uncertainties (e.g. Sims et al. 2025a,b; Liu et al. 2025; Tutt et al. 2026) and preliminary results of that analysis will be discussed in an upcoming technical memo. Early indications from the present work already suggest that the impact of beam uncertainty on power-spectrum inference is non-negligible, and that the sensitivity of inference to beam uncertainty depends on the level of foreground contamination in the data.

The above demonstrator provides the starting point for a broader programme. This will include future, more detailed validation analyses that expand on the demonstrator study, using higher fidelity simulations of the interferometric data (e.g. Cavillot et al. 2020, 2024; Gueuning et al. 2022, 2025; O'Hara et al. 2024; Gueuning et al. 2026), applying the analysis to other radio telescopes and targeting publication in a series of peer-reviewed journal articles. Furthermore, it will develop validation capability that can support SKA and precursor science across multiple workflows, levels of realism, and analysis products. In this memo, we use the term "ValSKA" to denote that broader framework-level ambition.

The remainder of this memo is organised as follows. Section 2 motivates validation as a scientific requirement for reliable scientific inference in precision astrophysics and cosmology applications[3] with SKA and precursor data sets. Section 3 defines ValSKA as a reusable framework for validation campaigns and analysis diagnostics, while Section 4 presents a conceptual view of its relationship to science drivers, shared infrastructure, case-specific modules, and validation outputs. Section 5 identifies design principles for developing ValSKA[4] as a modular, reproducible validation environment, and Section 6 outlines near-term priorities for moving from the current demonstrator towards a broader validation framework. Finally, Section 7 summarises the memo and describes plans for upcoming work.

2 Validation as a Scientific Requirement

> *"The first principle is that you must not fool yourself – and you are the easiest person to fool."*
> — Richard P. Feynman

For SKA and precursor scientific workflows, particularly in precision astrophysics and cosmology applications where analyses attempt to extract a faint signal from complex, instrument-coupled data, successful execution alone is not sufficient evidence of scientific trustworthiness. The relevant question is whether a workflow recovers the correct scientific answer for the correct reasons, with quantified uncertainty, under observing and modelling assumptions that are understood. This is a substantially stronger requirement than basic software correctness.

Modern cosmological experiments operate in a regime where systematic errors from imperfect modelling of instrumental response, calibration effects, foreground emission, and other astrophysical contributions can dominate the cosmological error budget. This is particularly true in low-frequency radio cosmology, where measurements of the cosmological signal are challenged by foreground contamination exceeding the target signal by several orders of magnitude. In such regimes, traditional notions of verification, in the sense of implementation correctness, must be augmented with validation analyses that ascertain the extent to which scientific conclusions can be trusted under realistic uncertainties in the instrument and sky model used for calibration and analysis of the data.

[3] `https://www.skao.int/en/explore/science-goals`
[4] `https://github.com/uksrc/ValSKA-HERA-beam-FWHM`

Analyses of precursor and pathfinder data have already revealed the importance of validation in this setting. With signal detection limited by systematic effects rather than thermal noise, the scientific interpretation of the result depends on understanding how that result was produced. Pipeline studies have shown that weakly diagnosed analysis choices can suppress cosmological signal or underestimate uncertainty (Cheng et al., 2018), while imperfect characterisation and calibration of the interferometer can introduce systematic effects that motivate detailed simulation tests. Conversely, modular and end-to-end validation studies have shown how synthetic data, controlled perturbations, and explicit scope statements can build confidence in a pipeline's behaviour under stated assumptions (Aguirre et al., 2022).

The need for such validation follows from the structure of modern radio astronomy analysis pipelines. Inference commonly depends on a chain of coupled elements: sky models, instrument models, simulators, calibration assumptions, statistical machinery, and post-processing choices. As a result, failures can arise in several ways:

- a processing step may contain fragile assumptions that may be violated by the data;
- imperfect instrumental characterisation and calibration can lead to unmodelled systematics, resulting in biased astrophysical and cosmological inference or underestimated uncertainty;
- a model may fit an aggregate data product while misattributing signal between components;
- overfitting or filtering of the data can lead to signal loss and reduced sensitivity to the target signal.

Validation is a scientific activity situated between software engineering, statistical inference, and instrument understanding. In practical terms, the validation objective is to determine whether an inference workflow is trustworthy under stated conditions and to identify the instrument and sky characterisation requirements for that trustworthiness. This is a stronger requirement than software correctness, which is necessary but not sufficient for scientific trustworthiness.

A validation claim is necessarily conditional on scope. A useful validation exercise should therefore state which workflow components are being validated, which sources of instrument, sky, noise, modelling, or algorithmic uncertainty are included, which metrics are used to judge performance, and which assumptions remain outside the tested domain. This conditional framing is important: validation provides evidence for reliability under stated conditions; it does not prove universal correctness.

The relevant validation modes depend on the scientific question. For ValSKA, important modes include:

- component-level tests of scientifically relevant behaviour;
- end-to-end synthetic-data studies with known injected signals;
- null tests;
- systematic-uncertainty propagation;
- inference-robustness studies under plausible variations in forward models.
- cross-pipeline or cross-model comparisons;

Validation metrics will likewise be case-dependent, but commonly include:

- false-detection behaviour;
- sensitivity to instrument or sky-model perturbations;
- signal recovery, signal loss or bias.

In the UKSRC setting, validation outputs target more than one audience. They should inform scientists deciding whether a workflow is fit for a given analysis and instrument or calibration teams seeking quantitative guidance on tolerable uncertainties. They should also support external reuse by making the basis for trust explicit rather than implicit.

3 ValSKA as a Framework

Validation has emerged as a central methodological component across multiple areas of astronomy (see Appendix A). The present ValSKA work begins with a demonstrator science case. Its longer-term objective is to establish a unified validation framework that can be applied to multiple experiments and science cases together with software infrastructure that makes validation studies easier to define, execute, inspect, compare, and communicate.

This objective has both scientific and operational components. Scientifically, the framework must support tests that reveal observational setups or modelling assumptions that lead to spurious or biased signal recovery. Operationally, it must make those tests reproducible, auditable, and transferable across campaigns and contributors. At a high level, ValSKA plans to support the following classes of activity:

- controlled, simulation-based studies of known systematic effects;
- model validation in Bayesian settings, including evidence checks and related null-test logic;
- parameter-recovery and bias-diagnosis workflows under perturbations to instrument and sky assumptions;
- interfaces to downstream science products, such as recovered power spectra, posterior summaries, image-domain products, and benchmark tables;
- reproducible orchestration across local and HPC environments;
- provenance-rich artefacts that can be audited, shared, and cited.

As a framework, ValSKA will make explicit its conventions, interfaces, diagnostics, assumptions, and mechanisms for communicating trust. It should support both modular tests of individual components and more integrated end-to-end studies of full workflow behaviour, following the pattern described by Aguirre et al. (2022).

4 Conceptual View of ValSKA

Figure 1 shows a conceptual view of the planned ValSKA framework, illustrating how it connects science drivers and validation objectives to reusable shared infrastructure, case-specific workflow modules, validation metrics, and decision-relevant outputs. This framework architecture makes provenance capture, reporting, artefact publication, and feedback into future science and instrument priorities explicit parts of the same validation system.

The figure organises ValSKA around four linked regions of activity, supported by a cross-cutting shared framework:

Drivers and Scope. This region gathers the scientific and instrumental drivers for a validation campaign, including UK science priorities, calibration questions, real data products, sky and instrument models, simulators, and benchmark assets.

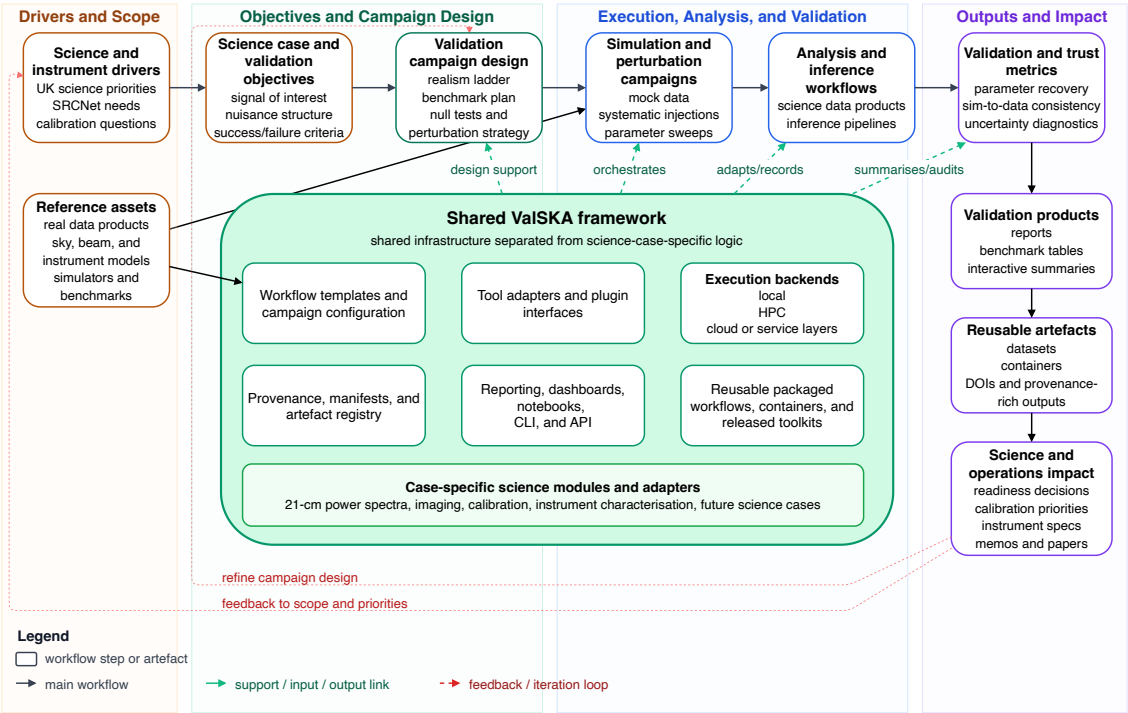

Figure 1: Schematic representation of the iterative ValSKA validation framework combining a left-to-right validation workflow with a shared framework core. Broadly, science drivers and validation objectives feed into campaign design, simulation and perturbation studies, analysis and inference workflows, and trust metrics. A shared ValSKA layer provides common infrastructure for configuration, orchestration, tool adapters, provenance capture, reporting, and reusable packaged workflows, while case-specific modules remain explicit. Outputs include validation products, reusable artefacts, and science or operations impacts, with feedback loops informing future scope and campaign design.

Objectives and Campaign Design. This region turns those inputs into explicit science-case and validation objectives, success or failure criteria, benchmark plans, null tests, and perturbation strategies.

Execution, Analysis, and Validation. This region captures the main reproducible workflow: simulation and perturbation campaigns feed analysis and inference workflows, which are assessed through validation and trust metrics such as parameter recovery, simulation-to-data consistency, and uncertainty diagnostics.

Outputs and Impact. This region records the products that make validation useful beyond a single run, including reports, benchmark tables, interactive summaries, datasets, containers, provenance-rich outputs, readiness decisions, calibration priorities, instrument specifications, memos, and papers.

Shared ValSKA Framework. Across the central part of the workflow, the shared framework provides reusable infrastructure for workflow templates, campaign configuration, tool adapters, execution backends, provenance manifests, artefact registries, reporting interfaces, packaged workflows, released toolkits, and case-specific science modules.

Taken together, these regions define ValSKA as a validation environment in which workflows can be designed, executed, tested, interpreted, and fed back into future science and instrument priorities in a controlled manner.

5 Design Considerations

The ValSKA framework defined in Sections 3 and 4 is most useful when accompanied by concrete design considerations. Here we identify several considerations that are important for the current stage of the programme.

5.1 Validation products should be treated as scientific artefacts

If validation is to influence scientific interpretation, then its outputs must be interpretable, documented, and recoverable. Reports, benchmark datasets, diagnostic summaries, and campaign manifests are scientific artefacts in their own right.

5.2 Shared infrastructure should remain distinct from science-case logic

Different use cases may be expected to require different simulations, inference engines, likelihoods, and diagnostics. For that reason, we favour modularity in the design and implementation of the ValSKA framework. Shared orchestration, provenance handling, and reporting are initial candidates identified for common infrastructure; science-specific models should remain explicit and replaceable.

5.3 Reproducibility should be a default property

Run configuration, ValSKA version, and generated artefacts should be recoverable after the fact. A result that cannot be re-run or audited has limited value as validation evidence.

5.4 Diagnostics should be prioritised over favourable demonstrations

Validation is most useful when it identifies where a method fails, what biases dominate, and which assumptions matter most. Negative outcomes (e.g. spurious detections and biased inferences) are scientifically valuable when they clarify the operating boundaries of a method.

5.5 Scientific realism should increase in stages

Early studies can productively use simplified scenarios when those isolate a specific effect. More advanced studies will move toward higher-fidelity simulations, parameter recovery, and simulation-to-data comparison under progressively more realistic conditions.

6 Near-Term Programme

Our first priority is to strengthen the path from the current demonstrator repository toward a more durable validation framework. Practical priorities include:

1. Defining scope: clarifying the conceptual scope of ValSKA as a framework for validation campaigns (this memo);
2. Usability: continuing to improve reproducible orchestration and tooling around existing case-study campaigns;
3. Modularity: defining boundaries between case-specific code and potential shared infrastructure;
4. Shareability: improving the packaging and documentation of validation outputs for sharing and later reuse;
5. Communicating progress: using technical memos to record methodology, benchmark results, and architectural intent as the project grows.

These priorities align naturally with a staged science-tooling programme: iterative validation workflows in the short term, followed by more production-ready and interoperable tooling over time.

7 Conclusion

ValSKA is the beginning of a broader validation effort for SKA and precursor science. By combining scientific rigour, reproducible execution, diagnostic depth, and reusable artefacts, it will provide a shared open-source validation workflow and practical infrastructure to support applying those methods to UKSRC target science cases and to SKA and precursor analysis problems more broadly.

A further goal is for ValSKA to provide a practical route for UKSRC science-case experts and other UK teams to contribute to and reuse validation infrastructure. Case-specific modules can encode domain expertise, while shared ValSKA conventions, orchestration, provenance, and reporting make those contributions inspectable, reusable, and transferable across workflows.

This memo is the first in a planned technical memo series that will contribute to that goal by providing a record of validation methods, benchmark results, architectural decisions, and known limitations. Used in this way, the memo series becomes a practical companion to the software itself.

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A Validation Exemplars in Astronomy and Cosmology

This appendix provides a more detailed discussion of validation strategies in the context of a selection of precision astrophysics pipelines. It is intended to complement the more general discussion in the main text with specific examples and reference implementations from the literature. The examples are not intended to be exhaustive, but rather to illustrate the diversity of validation approaches and their relevance to the challenges of SKA and precursor science. Examples include:

- the Planck Collaboration's extensive use of validation workflows via null tests, cross-frequency and detector-split consistency checks, and end-to-end simulations, with parameter robustness explicitly quantified under alternative likelihood constructions (e.g. Planck Collaboration et al. 2016);
- the practices underpinning gravitational-wave detection, where significance is established via injections, background estimation, and multi-detector coincidence (Abbott et al. 2016);

- the standard use of image simulations and internal consistency tests in large-scale structure and weak-lensing surveys (e.g. Troxel et al. 2018);
- the Event Horizon Telescope (EHT) collaboration's use of independent imaging pipelines, synthetic-data challenges, and cross-validation against geometric and GRMHD forward models to establish robustness of image-domain features (Event Horizon Telescope Collaboration et al. 2019a,b);
- the increasing formalisation of validation as a primary deliverable in 21-cm cosmology experiments, with independent pipelines employed to expose algorithmic sensitivities (e.g. Jacobs et al. 2016), instrumental and modelling uncertainties propagated through the analysis chain to assess their impact on recovered spectra (e.g. Monsalve et al. 2017; Sims & Pober 2020; Mertens et al. 2020; Liu et al. 2025; Tutt et al. 2026), and end-to-end synthetic pipelines used to quantify signal loss and bias (e.g. Barry et al. 2016; Aguirre et al. 2022; Abdurashidova et al. 2022).

These examples demonstrate the role of validation in the interpretation of complex data, and that while it can be implemented in a variety of ways depending on the scientific context, many cases share common features: the use of synthetic data, consistency checks including null tests, and controlled perturbations to understand the limits of a pipeline's validity or build confidence in its behaviour under stated assumptions.